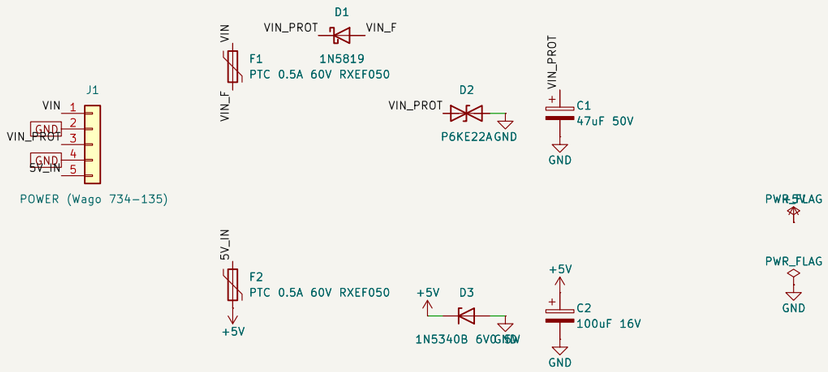
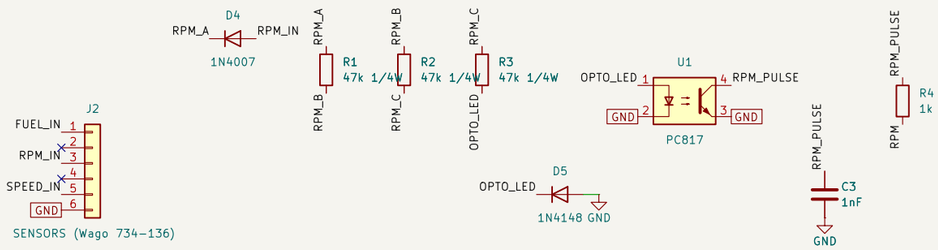
POWER INPUT AND PROTECTION



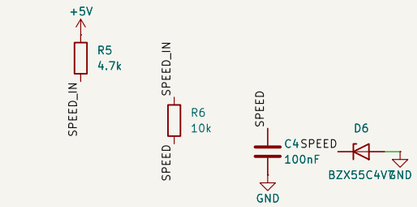
Set the external step-down module to 5.0 V BEFORE connecting it. If its preset fails, D3 conducts and F2 trips.
P6KE22A clamps load-dump spikes (Vc ~31 V) below the 40 V limit of LM2596 modules. D1 blocks reverse polarity.

RPM INPUT – OPTOCOUPLER (ignition coil)

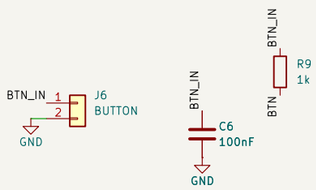


Coil primary pulses are 200–400 V (more with the spark plug disconnected).
Never run the engine with the plug cap off. HV nets (RPM_IN..RPM_C) keep 2.5 mm clearance on the PCB.
Firmware: PA6 as TIM3_CH1 input capture WITH internal pull-up; ignore pulses closer than ~2 ms (ignition ringing).

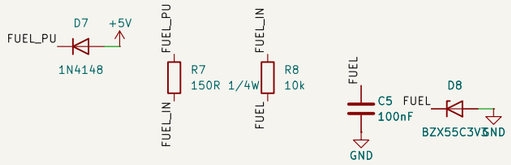
WHEEL SPEED INPUT



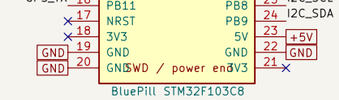
PUSH BUTTON



FUEL LEVEL INPUT

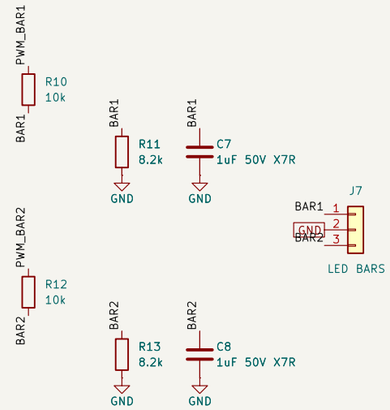


MCU – STM32 BLUE PILL

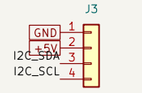


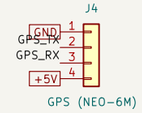
Pin numbers follow the Blue Pill headers: 1–20 = VB..GND row, 21–40 = 3.3/GND/5V..B12 row.
Pin 21 (3.3) is tied to pin 18 inside the module. Check the silkscreen labels against the module before soldering.
PA0 (button) and PA6 (RPM) need the internal pull-up enabled in firmware.

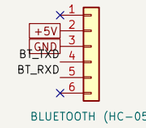
LED BAR GRAPH DRIVE (PWM -> DC)



CONNECTORS







Single-sided, through-hole only (home-made PCB)
Optocoupled RPM input, protected power and sensor inputs

Sheet: /
File: control.kicad_sch

Title: Ordenador de abordo – Control board v2

Size: A3	Date: 2026-10-14	**Rev: 2.0**
KiCad E.D.A. 10.0.3		Id: 1/1